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Technology of Metallic Materials A.Y 2025-26

Mechanical Engineering Course

Aluminum alloys



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DISAT

Lecture n.14

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Aluminum

- Melting point 660°C – softens at 150-200°C
- FCC crystal structure – no phase transformation
- No ductile-to-brittle transition
- Density = 2.7 gm/cm³ (vs. 7.7 gm/cm³ for Fe)
- E = 70 GPa (vs. 210 GPa for Fe)
- Cost in sheet form: ~4.5 x mild steel
(~1.5 x for equal thickness)
- Pure Al Y.S. ~35 MPa (alloys up to 560 MPa)
- High electrical conductivity (62% IASC*)
- Easily formed by extrusion, forging rolling etc (foil), can be formed at low temperatures and intricate shapes (FCC crystal structure)

*International Annealed Standard for Copper

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General Aluminum alloys

- Good **machining** properties
- Alloys are **weldable** (inert gas).
- **Fatigue strength** low, especially in pptn – hardened alloys
- **Corrosion resistance** excellent – thin strong oxide coating. Can anodise.
- Pptn- hardened alloys can suffer **stress-corrosion cracking** (ppt-free zones near grain boundaries)
- Good electrical and heat **conductor**.

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American designation

Casting and Wrought Aluminum alloys Designation of composition with four digits which specify elements.

Alloy Series	Principal elements wrought	Alloy Series	Principal elements casting
1xxx	Min 99.000% Al	1xx.x	99.000% min Al
2xxx	Cu	2xx.x	Cu
3xxx	Mn	3xx.x	Si + Cu and/or Mg
4xxx	Si	4xx.x	Si
5xxx	Mg	5xx.x	Mg
6xxx	Mg and Si	6xx.x	Unused Series
7xxx	Zn	7xx.x	Zn
8xxx	other elements	8xx.x	Sn
		9xx.x	Other Elements

<http://www.alcotec.com/us/en/solutions/-Understanding-the-Aluminum-Alloy-Designation-System.cfm>

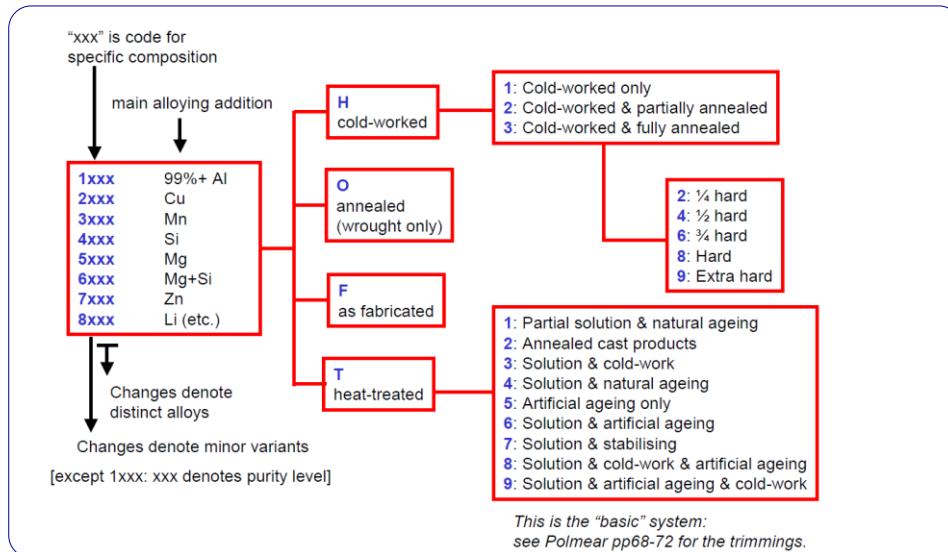
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Classification of Wrought/Cast Alloys and Heat Treatment designation



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Effect of Alloying Elements

- Cu: High strength from age hardening, not weldable (AA2xxx)
- Mn: Solution hardening (AA3xxx)
- Mn+Mg: Solution hardening even better
- Mg: Solution hardening – best corrosion resistance next to pure Al (AA5xxx)
- Mg+Si: Moderately heat treatable and corrosion resistance and weldability – good medium alloy (AA6xxx)
- Zn+Mg: High strength, not weldable (AA7xxx)
- Alclad
 - High purity Al skin for corrosion resistance (Passivated natural coating)
 - May protect galvanically

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Al-alloys strengthening

- Strengthening mechanisms
 - solid solution hardening
 - cold work (strain hardening)
 - age hardening $\sigma_{ys} = \sigma_o + kd^{-1/2}$
 - Age hardening
 - type of phase diagram and aging behaviour
 - Alloy designation (previous slide)
 - Temper designation
 - O, F, H_{1x} (x=2,4,6,8), T3, T4, T6, T8
- Fully annealed

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Alloying Elements Promoting Strengthening by Aging H.T.

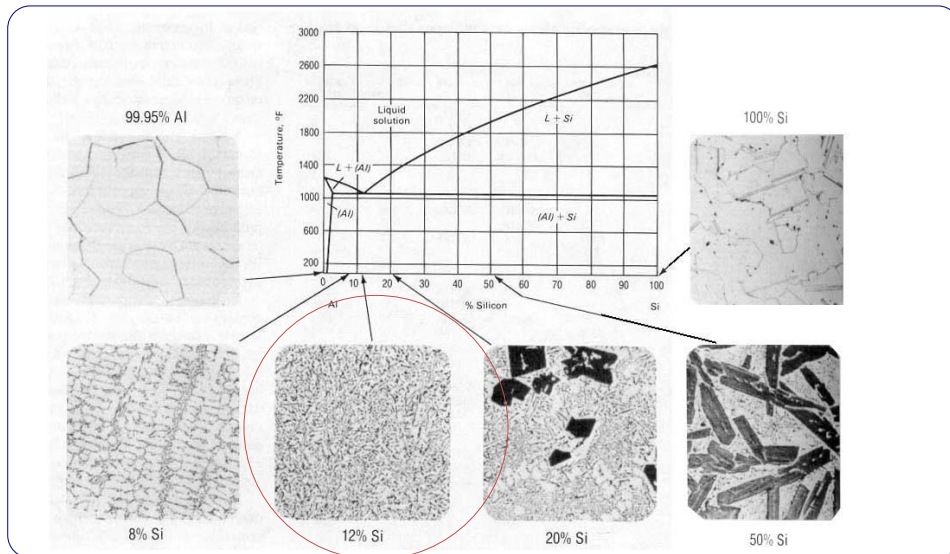
	non-heat treatables	heat treatables
Al>99.0%	1xxx (foil)	
Copper		2xxx+Mg+Mn (aircraft)
Manganese	3xxx +Si	↑ ↑
Magnesium	5xxx +Mn+Cr	↓ ↓ ↓
Mg+Si		6xxx+Cu+Si+Cr+Mn
Zinc		7xxx+Mg ↓
Other elements		8xxx (Li-aircraft)

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Phase diagram of casting Al-Si alloys



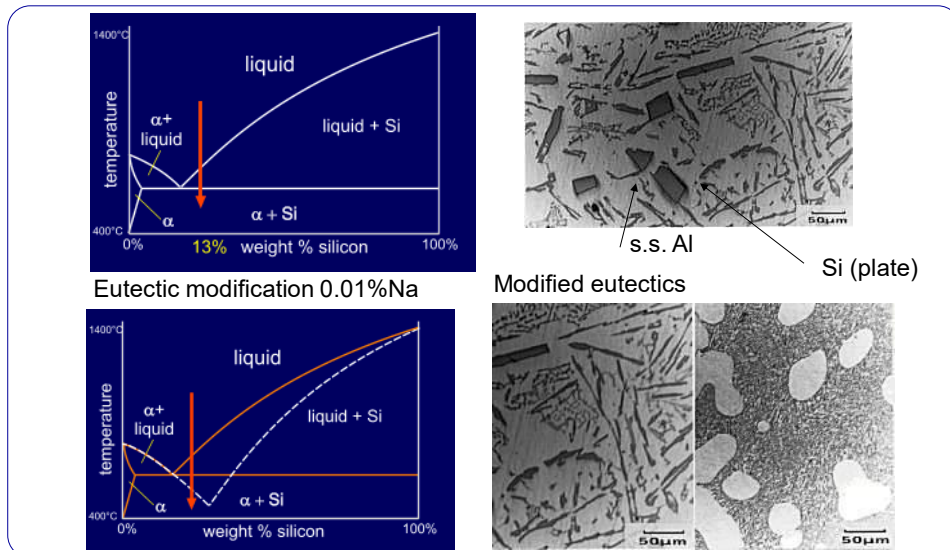
Note: changes of microstructure with increasing %Si in Al matrix.

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Toughening of Al-Si Alloys: eutectics modification without eutectics modifier

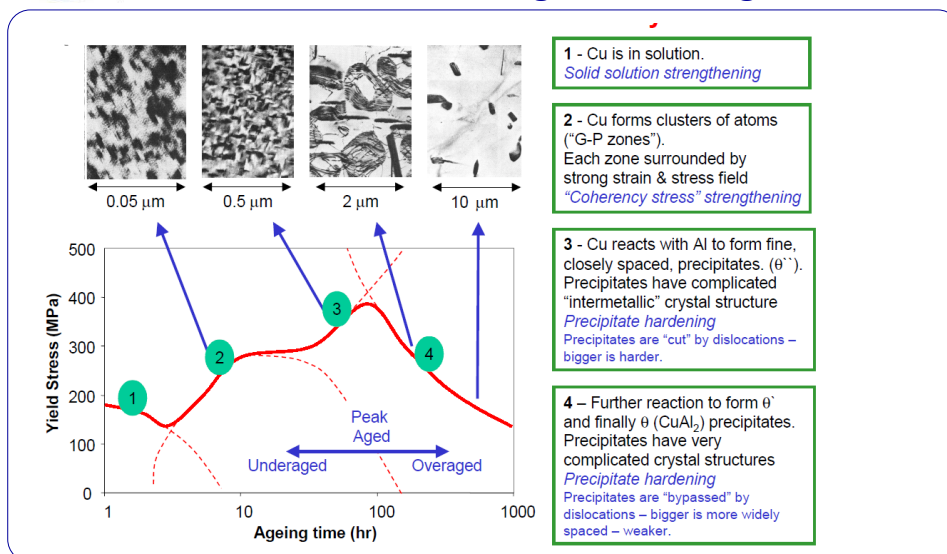


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The 2xxx Al-Cu series alloys: age hardening



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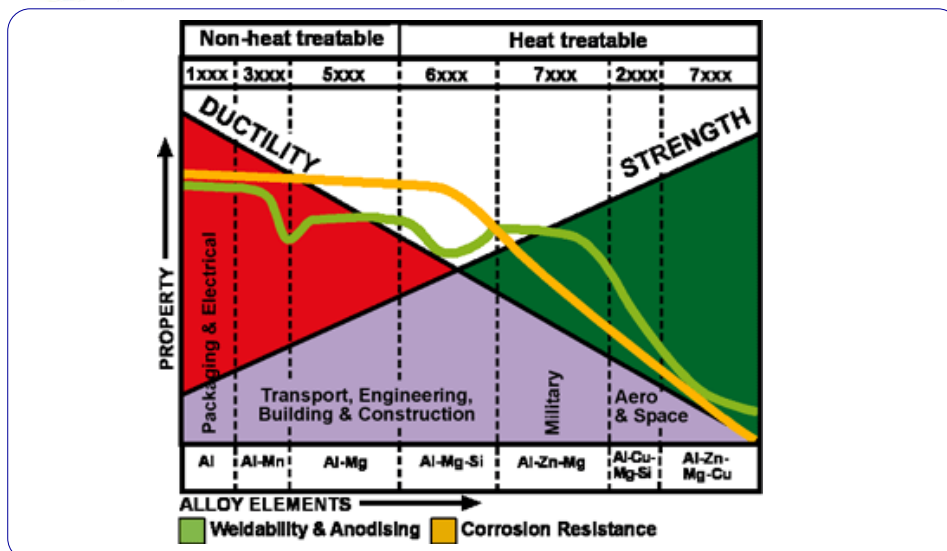
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SUMMARY

Ductility, Tempering, Weldability Corrosion resistance, Applications



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Commercial Wrought/Cast Alloy Classes and Applications

Alloy series	Composition	σ_y (MPa)	UTS (MPa)	ϵ_F (%)	Notes
1000	"pure" Al.	30 - 100	100 - 135	up to 50	Foil, decoration, electrical conductors
2000	~4.5% Cu (+Mn, Si, Mg) age-hardened	up to 480	up to 520	5 - 20	General purpose forgings and extrusions, esp. airframes.
3000	~1% Mn, Mg. cold-worked	up to 215	up to 290	5	Ductile sheet for cladding trucks, trailers. Food containers. Drink cans.
4000	12% Si, (+ Mg, Ni, Cu) forgeable, age-hardened	~295	~325	0.5	IC engine pistons. (aka LM13, A332)
5000	~5% Mg (+Cr, Mn). cold-worked	up to 350	up to 415	15	Good formability & weldability. Excellent corrosion resistance. Structural applications, esp. marine.
6000	up to 1% Mg-Si. age-hardened	~275	~310	12	Hot extrusions. Window frames, etc.
7000	up to 8% Zn (+Mg, Cu, Cr). age-hardened	500	575	11	Highest strength alloys. Aircraft structures.
8000	"Other", e.g. up to 2.5% Li				Novel and specialist alloys
Cast	near Al-13%Si eutectic + 0.01%Na	up to 200	up to 300	2 - 5	Automotive castings. Can age-harden if Cu and Mg added.

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